

An Introduction to Sturm–Liouville Theory

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1. Regular Sturm–Liouville problems

Let $a < b$ and let $p, q, w \in C([a, b])$ satisfy $p(x) > 0$ and $w(x) > 0$ on $[a, b]$. The (formal) Sturm–Liouville differential expression is

$$L[y] := -(p(x)y'(x))' + q(x)y(x). \quad (1.1)$$

A *regular Sturm–Liouville problem* on $[a, b]$ asks for nontrivial solutions of

$$L[y] = \lambda w(x) y, \quad a < x < b, \quad (1.2)$$

subject to separated, self-adjoint boundary conditions

$$\begin{aligned} y(a) \cos \alpha + p(a)y'(a) \sin \alpha &= 0, & 0 \leq \alpha < \pi, \\ y(b) \cos \beta + p(b)y'(b) \sin \beta &= 0, & 0 \leq \beta < \pi. \end{aligned} \quad (1.3)$$

We adopt the weighted inner product

$$\langle f, g \rangle_w := \int_a^b f(x) \overline{g(x)} w(x) dx. \quad (1.4)$$

Example 1.1. On $[0, \pi]$ with $p = w \equiv 1$ and $q \equiv 0$ and Dirichlet data $y(0) = y(\pi) = 0$, the eigenpairs are $\lambda_n = n^2$ and $y_n(x) = \sin(nx)$, $n = 1, 2, \dots$

1.1. Green’s formula and self-adjointness

Lemma 1.2 (Green’s formula). *For $u, v \in C^2[a, b]$,*

$$\int_a^b (v L[u] - u L[v]) dx = \left[p(x)(vu' - uv') \right]_{x=a}^{x=b}. \quad (1.5)$$

Proof. Using $L[u] = -(pu')' + qu$ and $L[v] = -(pv')' + qv$,

$$\begin{aligned} \int_a^b (vL[u] - uL[v]) dx &= \int_a^b (-v(pu')' + u(pv')') dx \\ &= \left[-vpu' + upv' \right]_a^b + \int_a^b (pv'u' - pu'v') dx \\ &= \left[p(vu' - uv') \right]_a^b. \end{aligned}$$

□

Proposition 1.3 (Symmetry). *If u and v satisfy the same self-adjoint separated boundary conditions (1.3), then the boundary term in (1.5) vanishes and*

$$\int_a^b v L[u] dx = \int_a^b u L[v] dx. \quad (1.6)$$

Proof. We show the bracket vanishes at a ; the argument at b is identical. If $\sin \alpha = 0$ then $y(a) = 0$ for any admissible y , so $v(a) = u(a) = 0$ and $p(a)(vu' - uv')|_{x=a} = 0$. If $\sin \alpha \neq 0$, then the boundary condition gives $y'(a) = -\frac{\cot \alpha}{p(a)} y(a)$. Hence for admissible u, v ,

$$p(a)(vu' - uv') \Big|_{x=a} = p(a) \left(v(a) \cdot \left(-\frac{\cot \alpha}{p(a)} u(a) \right) - u(a) \cdot \left(-\frac{\cot \alpha}{p(a)} v(a) \right) \right) = 0. \quad \square$$

Corollary 1.4 (Orthogonality of eigenfunctions). *If u, v are eigenfunctions with eigenvalues $\lambda \neq \mu$, then $\langle u, v \rangle_w = 0$.*

Proof. Apply Lemma 1.2 with $L[u] = \lambda wu$ and $L[v] = \mu wv$. The boundary term vanishes by Proposition 1.3, hence $(\lambda - \mu) \int_a^b uv w dx = 0$. □

1.2. Zeros are simple; uniqueness at zeros

Lemma 1.5 (Simplicity of zeros). *If y solves (1.2) and $y(x_0) = y'(x_0) = 0$ for some $x_0 \in [a, b]$, then $y \equiv 0$.*

Proof. This is the standard uniqueness theorem for linear second-order ODE initial value problems. □

Corollary 1.6. *Nontrivial solutions of (1.2) have only simple zeros: if $y(x_0) = 0$ then $y'(x_0) \neq 0$.*

2. Separation of zeros (Sturm separation)

We first state and prove the theorem in the classical Schrödinger form $y'' + Q(x)y = 0$ (the case $p = w \equiv 1$). The general case follows by the Liouville transform (Section 6).

Theorem 2.1 (Sturm separation, $p = w \equiv 1$). *Let y_1, y_2 be linearly independent solutions of $y'' + Q(x)y = 0$ on (a, b) with $Q \in C([a, b])$. Between any two consecutive zeros of y_1 there lies exactly one zero of y_2 , and conversely.*

Proof. Let $\alpha < \beta$ be consecutive zeros of y_1 ; by Corollary 1.6, the zeros are simple, so $y_1'(\alpha)$ and $y_1'(\beta)$ are nonzero and have opposite signs. The Wronskian $W := y_1 y_2' - y_1' y_2$ satisfies $W' = y_1 y_2'' - y_1'' y_2 = -Q y_1 y_2 + Q y_1 y_2 = 0$, so W is constant. Evaluate at α and β : $W(\alpha) = -y_1'(\alpha) y_2(\alpha)$, $W(\beta) = -y_1'(\beta) y_2(\beta)$. If y_2 had no zero in (α, β) , then $y_2(\alpha)$ and $y_2(\beta)$ would have the same sign (none of them can be zero as then the Wronskian would be zero, negating linear independence); since $y_1'(\alpha)$ and $y_1'(\beta)$ have opposite signs, $W(\alpha)$ and $W(\beta)$ would have opposite signs—contradiction to constancy of W . Thus y_2 has at least one zero in (α, β) . Simplicity of zeros rules out more than one. □

Remark 2.2. For the general self-adjoint form $-(py')' + qy = \lambda wy$, the Liouville transform of Section 6 reduces the equation to Schrödinger form on a reparametrized interval, and zeros are preserved under this monotone change of variable. Thus Theorem 2.1 holds in full generality.

3. Picone identity and comparison theorems

3.1. Picone identity

Lemma 3.1 (Picone identity, general self-adjoint form). *Let $p > 0$ be continuous on $[a, b]$ and let y_1, y_2 be nontrivial solutions of*

$$-(py_1')' + q_1 y_1 = 0, \quad -(py_2')' + q_2 y_2 = 0 \quad (3.1)$$

on an interval where y_2 has no zeros. Then

$$\left[p \left(\frac{y_1}{y_2} \right) (y_1' y_2 - y_1 y_2') \right]' = (q_2 - q_1) y_1^2 + p \left(y_1' - \frac{y_1}{y_2} y_2' \right)^2 \geq (q_2 - q_1) y_1^2. \quad (3.2)$$

Proof. Set $\Phi := p \frac{y_1}{y_2} (y_1' y_2 - y_1 y_2')$. Differentiate:

$$\begin{aligned} \Phi' &= p' \frac{y_1}{y_2} (y_1' y_2 - y_1 y_2') + p \frac{y_1'}{y_2} (y_1' y_2 - y_1 y_2') + p \frac{y_1}{y_2} (y_1'' y_2 + y_1' y_2' - y_1' y_2' - y_1 y_2'') \\ &\quad - p \frac{y_1}{y_2^2} y_2' (y_1' y_2 - y_1 y_2') \\ &= p \left(y_1' - \frac{y_1}{y_2} y_2' \right)^2 + \frac{y_1}{y_2} (p y_1'' y_2 - p y_1 y_2'') + \frac{p'}{y_2} y_1 (y_1' y_2 - y_1 y_2') - \frac{p y_2'}{y_2^2} y_1 (y_1' y_2 - y_1 y_2'). \end{aligned}$$

Rewrite the last two terms using $p' y_1' y_2 - p' y_1 y_2' - p y_2' y_1' + p y_2' \frac{y_1}{y_2} y_2' = (p y_1')' y_2 - (p y_2')' y_1$. Hence

$$\begin{aligned} \Phi' &= p \left(y_1' - \frac{y_1}{y_2} y_2' \right)^2 + \frac{y_1}{y_2} \left((p y_1')' y_2 - (p y_2')' y_1 \right) \\ &= p \left(y_1' - \frac{y_1}{y_2} y_2' \right)^2 + \frac{y_1}{y_2} \left((q_1 y_1) y_2 - (q_2 y_2) y_1 \right) \\ &= p \left(y_1' - \frac{y_1}{y_2} y_2' \right)^2 + (q_2 - q_1) y_1^2, \end{aligned}$$

as claimed. $\square$

3.2. Sturm comparison (constant $p = w = 1$)

Theorem 3.2 (Sturm comparison). *Let $q_1, q_2 \in C([a, b])$ with $q_2 \geq q_1$ and not identical on any subinterval. If y_1 solves $y_1'' + q_1 y_1 = 0$ and has consecutive zeros at $\alpha < \beta$, then every nontrivial solution y_2 of $y_2'' + q_2 y_2 = 0$ has at least one zero in (α, β) . If $q_2 > q_1$ on (α, β) , then exactly one zero occurs.*

Proof. Assume y_2 has no zero on (α, β) ; multiply by -1 if needed to take $y_2 > 0$ there. Apply Lemma 3.1 with $p \equiv 1$ to obtain $\Phi' \geq (q_2 - q_1) y_1^2$. Integrate on $[\alpha, \beta]$. Since $y_1(\alpha) = y_1(\beta) = 0$, the boundary term $\Phi(\beta) - \Phi(\alpha) = 0$. Thus $0 \geq \int_{\alpha}^{\beta} (q_2 - q_1) y_1^2 dx$, impossible if $q_2 \geq q_1$ with strict inequality somewhere and $y_1 \not\equiv 0$. If $q_2 > q_1$ on (α, β) , separation of zeros (Theorem 2.1) excludes more than one zero. $\square$

3.3. Sturm–Picone comparison (general p, w)

Theorem 3.3 (Sturm–Picone comparison). *Let $p \in C([a, b])$ with $p > 0$ and $q_1, q_2 \in C([a, b])$. Suppose y_1 is a nontrivial solution of*

$$-(py_1')' + q_1 y_1 = 0$$

with consecutive zeros at $\alpha < \beta$ (so $y_1(\alpha) = y_1(\beta) = 0$ and $y_1 \neq 0$ on (α, β)). Let y_2 be any nontrivial solution of

$$-(py_2')' + q_2 y_2 = 0.$$

If $q_2 \geq q_1$ on (α, β) and $q_2 \not\equiv q_1$ on any subinterval of (α, β) , then y_2 has at least one zero in (α, β) .

Proof. If y_2 has a zero in (α, β) there is nothing to prove. Otherwise y_2 keeps a fixed sign on (α, β) ; multiplying by -1 if necessary, assume $y_2 > 0$ there. On any subinterval of (α, β) , Lemma 3.1 (Picone identity) yields

$$\left[p \left(\frac{y_1}{y_2} \right) (y_1' y_2 - y_1 y_2') \right]' = (q_2 - q_1) y_1^2 + p \left(y_1' - \frac{y_1}{y_2} y_2' \right)^2 \geq (q_2 - q_1) y_1^2.$$

Integrate from α to β . Since $y_1(\alpha) = y_1(\beta) = 0$, the boundary term vanishes:

$$\left[p \left(\frac{y_1}{y_2} \right) (y_1' y_2 - y_1 y_2') \right]_{\alpha}^{\beta} = 0,$$

hence

$$0 \geq \int_{\alpha}^{\beta} (q_2 - q_1) y_1(x)^2 dx.$$

By hypothesis $q_2 \geq q_1$ with strict inequality somewhere on (α, β) and $y_1 \neq 0$, so the integral is positive—a contradiction. Therefore y_2 must have at least one zero in (α, β) . $\square$

Remark 3.4 (About zeros at the endpoints). The theorem asserts a zero in the *open* interval (α, β) . If $y_2(\alpha) = 0$ or $y_2(\beta) = 0$, that still does not preclude an interior zero; the proof by contradiction assumes y_2 has no zero in (α, β) , not on $[\alpha, \beta]$.

Lemma 3.5 (Wronskian derivative for two Sturm–Liouville equations). *Let $p > 0$ on $[a, b]$ with $p \in C^1([a, b])$ (Lipschitz suffices), and $q_1, q_2 \in C([a, b])$. Suppose y_1, y_2 solve*

$$-(py_1')' + q_1 y_1 = 0, \quad -(py_2')' + q_2 y_2 = 0.$$

Define the p -weighted Wronskian $W := p(y_1 y_2' - y_1' y_2)$. Then

$$W'(x) = (q_2(x) - q_1(x)) y_1(x) y_2(x) \quad \text{for all } x \in (a, b). \quad (3.3)$$

Proof. Differentiate W and use $(py_1')' = q_1 y_1$, $(py_2')' = q_2 y_2$:

$$\begin{aligned} W' &= p'(y_1 y_2' - y_1' y_2) + p(y_1' y_2' + y_1 y_2'' - y_1'' y_2 - y_1' y_2') \\ &= p y_1 y_2'' - p y_1'' y_2 + p'(y_1 y_2' - y_1' y_2) \\ &= y_1 (p y_2'' + p' y_2') - (p y_1'' + p' y_1') y_2 = q_2 y_1 y_2 - q_1 y_1 y_2, \end{aligned}$$

which is (3.3). $\square$

Theorem 3.6 (Quotient monotonicity (Sturm–Picone in ratio form)). *Let $p \in C([a, b])$ with $p > 0$, and $q_1, q_2 \in C([a, b])$. Suppose y_1, y_2 are nontrivial solutions of*

$$-(py'_i)' + q_i y_i = 0 \quad \text{on } (a, b), \quad i = 1, 2.$$

Let $I = (\alpha, \beta) \subset (a, b)$ be an open interval on which y_2 has no zero (so y_2 keeps a fixed sign), and on which y_1 also has a fixed sign. Define $r := y_1/y_2$ on I . Then the identity

$$(p y_2^2 r')' = (q_1 - q_2) y_1 y_2 = (q_1 - q_2) r y_2^2 \quad (3.4)$$

holds pointwise on I . In particular:

- (i) *If $q_2 \geq q_1$ on I and $y_1 y_2 \geq 0$ on I (e.g. both $y_1, y_2 > 0$), then r is nonincreasing on I . If $q_2 > q_1$ on a set of positive measure in I , then r is strictly decreasing.*
- (ii) *If $q_2 \leq q_1$ on I and $y_1 y_2 \geq 0$ on I , then r is nondecreasing on I (strictly if $q_2 < q_1$ on a set of positive measure).*

Proof. On I we can write $r = y_1/y_2$ and compute

$$r' = \frac{y'_1 y_2 - y_1 y'_2}{y_2^2} \implies p y_2^2 r' = p (y'_1 y_2 - y_1 y'_2).$$

Differentiate and use the product rule:

$$(p y_2^2 r')' = (p y'_1)' y_2 + p y'_1 y'_2 - (p y_1)' y'_2 - p y_1 y''_2.$$

Since $(p y_1)' = p' y_1 + p y'_1$, the middle terms cancel, leaving

$$(p y_2^2 r')' = (p y'_1)' y_2 - y_1 (p y'_2)'$$

Now use the equations $-(p y'_i)' + q_i y_i = 0$, i.e. $(p y'_i)' = q_i y_i$, to get

$$(p y_2^2 r')' = q_1 y_1 y_2 - q_2 y_1 y_2 = (q_1 - q_2) y_1 y_2,$$

which is (3.4). If $q_2 \geq q_1$ and $y_1 y_2 \geq 0$ on I , then the right-hand side is ≤ 0 , hence $p y_2^2 r'$ is nonincreasing; since $p > 0$ and $y_2^2 > 0$ on I , it follows that r' does not change sign and is ≤ 0 . The strict case is identical when the right-hand side is < 0 on a set of positive measure. The nondecreasing variant is the same with the inequalities reversed. $\square$

Corollary 3.7 (Model comparison in Schrödinger form). *Let $K_1, K_2 \in C([0, T])$, and let y_i solve*

$$y''_i + K_i(t) y_i = 0, \quad y_i(0) = 0, \quad y'_i(0) = 1, \quad i = 1, 2.$$

Then there exists $t_0 \in (0, T]$ such that $y_i(t) > 0$ on $(0, t_0)$, and on that interval the quotient

$\frac{y_1}{y_2}$ is nonincreasing if $K_1 \geq K_2$ (strictly decreasing if $K_1 > K_2$ on a set of positive measure).

In particular, if $K_1 \geq K_2$ then $0 < y_1(t) \leq y_2(t)$ for $t \in (0, t_0)$.

Proof. Near $t = 0$, $y_i(t) = t + O(t^3)$, so $y_i > 0$ on $(0, t_0)$ and $r'(0) = 0$. Apply Theorem 3.6 with $p \equiv 1$ and $q_i = -K_i$ (since $-(y'_i)' + q_i y_i = 0$ equals $y''_i + K_i y_i = 0$ for $q_i = -K_i$). Then $K_1 \geq K_2$ means $q_2 \geq q_1$, and $y_1 y_2 > 0$, so $(y_2^2 r')' \leq 0$. With $r'(0) = 0$ we conclude $r'(t) \leq 0$ for $t \in (0, t_0)$ (strict $<$ under strict $K_1 > K_2$), hence y_1/y_2 is (strictly) nonincreasing and $y_1 \leq y_2$. $\square$

Lemma 3.8 (Volterra form from $y'' = -Ky$). Let $K \in C([0, T])$ and let y solve

$$y''(t) = -K(t)y(t), \quad y(0) = 0, \quad y'(0) = 1.$$

Then for every $t \in [0, T]$,

$$y(t) = t - \int_0^t (t-s) K(s) y(s) ds. \quad (3.5)$$

More generally, if $y(0) = a$ and $y'(0) = b$, then

$$y(t) = a + bt - \int_0^t (t-s) K(s) y(s) ds. \quad (3.6)$$

Proof. Integrate $y'' = -Ky$ once on $[0, t]$ and use $y'(0) = 1$:

$$y'(t) - y'(0) = \int_0^t y''(s) ds \Rightarrow y'(t) = 1 - \int_0^t K(s) y(s) ds.$$

Integrate again and use $y(0) = 0$:

$$y(t) - y(0) = \int_0^t y'(\tau) d\tau = \int_0^t \left(1 - \int_0^\tau K(s) y(s) ds\right) d\tau = t - \int_0^t \int_0^\tau K(s) y(s) ds d\tau.$$

Since Ky is continuous, Fubini/Tonelli applies on the triangle $\{(s, \tau) : 0 \leq s \leq \tau \leq t\}$, giving the standard identity

$$\int_0^t \int_0^\tau g(s) ds d\tau = \int_0^t \left(\int_s^t d\tau\right) g(s) ds = \int_0^t (t-s) g(s) ds.$$

With $g(s) = K(s)y(s)$ this yields (3.5). The case (3.6) is identical, starting from $y'(t) = b - \int_0^t K(s)y(s) ds$ and $y(t) = a + \int_0^t y'(\tau) d\tau$. $\square$

Lemma 3.9 (Near-0 behavior). Let $K \in C([0, T])$ and let y solve $y'' + K(t)y = 0$ with $y(0) = 0$, $y'(0) = 1$. Set $M_0 := \sup_{[0, T]} |K| < \infty$. Then there exists

$$t_0 \in \left(0, \min\left\{T, \sqrt{\frac{3}{2M_0}}\right\}\right]$$

such that, for all $t \in [0, t_0]$,

$$|y(t) - t| \leq \frac{M_0}{3} t^3 \quad \text{and hence} \quad y(t) \geq t - \frac{M_0}{3} t^3 \geq \frac{t}{2} > 0 \quad \text{for } t \in (0, t_0].$$

In particular $y(t) = t + O(t^3)$ as $t \downarrow 0$.

Proof. Integrate twice to get the Volterra form

$$y(t) = t - \int_0^t (t-s) K(s) y(s) ds. \quad (*)$$

For $t > 0$ define

$$\phi(t) := \sup_{0 < s \leq t} \frac{|y(s)|}{s} \in (0, \infty),$$

which is finite because y is C^1 with $y'(0) = 1$ and hence $|y(s)| \leq 2s$ for s small. Using $(*)$ and $|K(s)| \leq M_0$,

$$|y(t) - t| \leq \int_0^t (t-s) |K(s)| |y(s)| ds \leq M_0 \phi(t) \int_0^t (t-s)s ds = \frac{M_0}{6} \phi(t) t^3.$$

Divide by $t > 0$:

$$\frac{|y(t)|}{t} \leq 1 + \frac{M_0}{6} \phi(t) t^2.$$

Now fix $t > 0$ and repeat the same estimate with t replaced by any $0 < s \leq t$ to get $\frac{|y(s)|}{s} \leq 1 + \frac{M_0}{6} \phi(t) s^2 \leq 1 + \frac{M_0}{6} \phi(t) t^2$. Taking the supremum over $0 < s \leq t$ yields

$$\phi(t) \leq 1 + \frac{M_0}{6} t^2 \phi(t).$$

Hence, whenever $\frac{M_0}{6} t^2 \leq \frac{1}{2}$ (i.e. $t \leq \sqrt{3/M_0}$),

$$\phi(t) \leq \frac{1}{1 - \frac{M_0}{6} t^2} \leq 2.$$

Plugging back gives

$$|y(t) - t| \leq \frac{M_0}{6} \cdot 2 t^3 = \frac{M_0}{3} t^3, \quad 0 \leq t \leq \sqrt{3/M_0}.$$

Finally, choose $t_0 \leq \sqrt{3/(2M_0)}$; then for $t \in (0, t_0]$,

$$y(t) \geq t - \frac{M_0}{3} t^3 \geq t \left(1 - \frac{M_0}{3} t_0^2\right) \geq t \left(1 - \frac{1}{2}\right) = \frac{t}{2} > 0.$$

This also shows $y(t) = t + O(t^3)$ as $t \downarrow 0$. □

Remark 3.10 (If K is C^1 at 0). Differentiating $y'' = -Ky$ gives $y''(0) = 0$ and $y^{(3)}(0) = -K(0)$, hence the Taylor expansion

$$y(t) = t - \frac{K(0)}{6} t^3 + O(t^4).$$

Remark 3.11 (If K is C^1 at 0). Differentiating the ODE gives $y''(0) = 0$ and $y^{(3)}(0) = -K(0)$, so

$$y(t) = t - \frac{K(0)}{6} t^3 + O(t^4).$$

Corollary 3.12 (Model comparison with space forms). *Let $k \in \mathbb{R}$ and let sn_k be the space-form function with $\text{sn}_k'' + k \text{sn}_k = 0$, $\text{sn}_k(0) = 0$, $\text{sn}_k'(0) = 1$. If $K(t) \geq k$ on $(0, T)$ and y solves $y'' + K(t)y = 0$, $y(0) = 0$, $y'(0) = 1$, then*

$$\frac{y(t)}{\text{sn}_k(t)} \text{ is nonincreasing on } (0, T), \quad 0 < y(t) \leq \text{sn}_k(t) \text{ until the first zero of } \text{sn}_k.$$

If $K \leq k$, then y/sn_k is nondecreasing (hence $y \geq \text{sn}_k$).

Remark 3.13 (How this feeds Bishop–Gromov). Along a geodesic, let $Y_1, \dots, Y_{n-1}$ be orthogonal normal Jacobi fields with $Y_i(0) = 0$, $|Y_i'(0)| = 1$. Writing $y_i(t) := |Y_i(t)|$, each y_i satisfies $y_i'' + K_i(t)y_i = 0$ for some sectional curvatures $K_i(t) = \text{sec}(\text{span}\{Y_i(t), \dot{\gamma}(t)\})$. If $\text{sec} \geq k$, then by Corollary 3.12, y_i/sn_k is nonincreasing; hence

$$\frac{j(t)}{\text{sn}_k^{n-1}(t)} = \frac{\prod_{i=1}^{n-1} y_i(t)}{\text{sn}_k^{n-1}(t)} \text{ is nonincreasing in } t,$$

which is the pointwise density comparison behind Bishop’s inequality. (For Ricci bounds one uses the scalar Riccati comparison for $m(t) = \sum u_i(t)$ instead; see §?? and the mean-curvature comparison discussion.)

4. Sturm oscillation

Setting. Let $p \in C^1([a, b])$, $q, r \in C([a, b])$ with $p > 0$ and $r > 0$ on $[a, b]$. Consider the regular Sturm–Liouville equation

$$-(py')' + qy = \lambda ry \quad \text{on } [a, b], \quad (4.1)$$

with separated boundary conditions

$$y(a) \cos \alpha + p(a)y'(a) \sin \alpha = 0, \quad y(b) \cos \beta + p(b)y'(b) \sin \beta = 0, \quad (4.2)$$

where $\alpha, \beta \in [0, \pi)$.

For each $\lambda \in \mathbb{R}$ let $y(\cdot, \lambda)$ be the *left-normalized solution* of (4.1) determined by

$$y(a, \lambda) = -\sin \alpha, \quad p(a)y'(a, \lambda) = \cos \alpha \quad (4.3)$$

(the initial data are λ -independent). This convention is chosen so that the Prüfer phase below satisfies $\theta(a, \lambda) = -\alpha$ exactly (see (4.6)).

Prüfer transform. Set

$$u := y, \quad v := py'.$$

Then (4.1) is equivalent to the first-order system

$$u' = \frac{v}{p}, \quad v' = (q - \lambda r)u. \quad (4.4)$$

For nontrivial solutions, define *Prüfer variables* (ρ, θ) by

$$u = \rho \sin \theta, \quad v = \rho \cos \theta, \quad \rho = \sqrt{u^2 + v^2} > 0. \quad (4.5)$$

We pick the branch so that

$$\theta(a, \lambda) = -\alpha \quad (4.6)$$

which matches the left boundary condition, since

$$y(a) \cos \alpha + p(a)y'(a) \sin \alpha = \rho(a) \sin(\theta(a) + \alpha) = 0.$$

Lemma 4.1 (Phase and amplitude equations). *With (4.5)–(4.6),*

$$\theta'(x, \lambda) = \frac{1}{p(x)} \cos^2 \theta(x, \lambda) + (\lambda r(x) - q(x)) \sin^2 \theta(x, \lambda). \quad (4.7)$$

At any x with $\theta(x, \lambda) \in \pi\mathbb{Z}$ one has $\theta'(x, \lambda) = 1/p(x) > 0$; hence all zeros of $y(\cdot, \lambda)$ are simple.

Proof. Differentiate $u = \rho \sin \theta$, $v = \rho \cos \theta$ and compare with (4.4):

$$\rho' \sin \theta + \rho \theta' \cos \theta = \frac{\rho \cos \theta}{p}, \quad \rho' \cos \theta - \rho \theta' \sin \theta = (q - \lambda r)\rho \sin \theta.$$

Eliminate ρ' to obtain (4.7). If $\sin \theta = 0$, then $\cos \theta = \pm 1$ and $\theta' = 1/p > 0$. $\square$

Lemma 4.2 (Parameter dependence). *The left-normalized solution $y(\cdot, \lambda)$ depends C^1 on λ . In particular $y_\lambda := \partial_\lambda y$ and y'_λ are continuous and satisfy*

$$-(py'_\lambda)' + qy_\lambda = \lambda r y_\lambda + r y, \quad y_\lambda(a, \lambda) = 0, \quad p(a)y'_\lambda(a, \lambda) = 0. \quad (4.8)$$

Lemma 4.3 (Lagrange identity for $(y, \partial_\lambda y)$). *Define*

$$J(x) := p(x)(y y'_\lambda - y' y_\lambda).$$

Then

$$J'(x) = -r(x) y(x, \lambda)^2 \leq 0, \quad J(a) = 0, \quad (4.9)$$

hence

$$J(x) = - \int_a^x r(t) y(t, \lambda)^2 dt. \quad (4.10)$$

Proof. Use $(py')' = qy - \lambda ry$ and $(py'_\lambda)' = qy_\lambda - \lambda ry_\lambda - ry$ from (4.1) and (4.8):

$$\begin{aligned} J' &= y(py'_\lambda)' - y_\lambda(py')' = y(qy_\lambda - \lambda ry_\lambda - ry) - y_\lambda(qy - \lambda ry) \\ &= -r y^2. \end{aligned}$$

Since the initial data (4.3) are λ -independent, $J(a) = 0$. □

Proposition 4.4 (Phase monotonicity in λ). *For each fixed $x \in (a, b]$,*

$$\partial_\lambda \theta(x, \lambda) = \frac{\int_a^x r(t) y(t, \lambda)^2 dt}{y(x, \lambda)^2 + (p(x)y'(x, \lambda))^2} > 0. \quad (4.11)$$

Proof. From $u = \rho \sin \theta$, $v = \rho \cos \theta$,

$$\partial_\lambda \theta = \frac{v \partial_\lambda u - u \partial_\lambda v}{u^2 + v^2} = \frac{p y' y_\lambda - y p y'_\lambda}{y^2 + (p y')^2} = - \frac{J}{\rho^2}.$$

Combine with (4.10) and $\rho^2 > 0$. □

Proposition 4.5 (Eigenvalues are right-end phase hits). *A number λ is an eigenvalue of (4.1)–(4.2) iff*

$$\theta(b, \lambda) \equiv -\beta \pmod{\pi}. \quad (4.12)$$

Moreover, $\lambda \mapsto \theta(b, \lambda)$ is strictly increasing and continuous, so the eigenvalues form a simple increasing sequence $(\lambda_n)_{n \geq 1}$ and

$$\theta(b, \lambda_{n+1}) - \theta(b, \lambda_n) = \pi \quad \text{for all } n \geq 1. \quad (4.13)$$

Proof. The right BC reads $0 = \rho(b) \sin(\theta(b) + \beta)$, yielding (4.12). Strict monotonicity is from (4.11). Eigenvalues occur at the successive hits of the grid $\{-\beta + k\pi\}$, hence the phase gap is π . □

Theorem 4.6 (Sturm oscillation). *Let $\phi_n(x) := y(x, \lambda_n)$ be the eigenfunction for the n -th eigenvalue λ_n . Then ϕ_n has exactly $n - 1$ simple zeros in (a, b) .*

Proof. Set $\theta_n(x) := \theta(x, \lambda_n)$. A point $x \in (a, b)$ is a zero of ϕ_n iff $\theta_n(x) \in \pi\mathbb{Z}$, and the zero is simple by Lemma 4.1.

First, ϕ_1 has no interior zeros (classical: the principal eigenfunction can be chosen strictly positive; for a direct ODE proof, one may invoke Sturm's comparison or the minimization of the Rayleigh quotient).

Now compare ϕ_n and ϕ_{n+1} . Define $d(x) := \theta_{n+1}(x) - \theta_n(x)$. Using (4.11),

$$d(x) = \int_{\lambda_n}^{\lambda_{n+1}} \partial_\lambda \theta(x, s) ds > 0 \quad (x > a), \quad d(a) = 0, \quad d(b) = \pi,$$

so d is continuous and strictly positive on $(a, b]$.

Let $\xi_1 < \dots < \xi_m$ be the interior zeros of ϕ_n (with $m = n - 1$ by induction). At each ξ_j we have $\theta_n(\xi_j) = k_j\pi$ and hence

$$\theta_{n+1}(\xi_j) = k_j\pi + d(\xi_j), \quad d(\xi_j) \in (0, \pi).$$

Between consecutive zeros $\xi_j < \xi_{j+1}$, θ_n increases by π (each zero is a strict upward crossing). Therefore

$$\sin \theta_{n+1}(\xi_j) = (-1)^{k_j} \sin d(\xi_j), \quad \sin \theta_{n+1}(\xi_{j+1}) = (-1)^{k_j+1} \sin d(\xi_{j+1}),$$

which have *opposite signs* since $\sin d(\cdot) > 0$. By continuity, $\sin \theta_{n+1}$ vanishes *at least once* in each interval (ξ_j, ξ_{j+1}) . A similar one-sided evaluation at a and b (using the boundary versions of the comparison argument) shows there is also at least one zero in (a, ξ_1) and in (ξ_m, b) . Hence ϕ_{n+1} has at least $m + 1 = n$ interior zeros.

On the other hand, by (4.13),

$$\theta_{n+1}(b) - \theta_{n+1}(a) = (\theta_n(b) - \theta_n(a)) + \pi.$$

Each interior zero corresponds to a crossing of a level $k\pi$; because all such crossings are strict and upward, increasing the end phase by exactly π can add *at most one* additional crossing over the whole interval. Thus ϕ_{n+1} has at most $m + 1 = n$ interior zeros. Combining the two bounds, ϕ_{n+1} has exactly n interior zeros, which completes the induction. $\square$

Corollary 4.7 (Interlacing). *If $\xi_1 < \dots < \xi_{n-1}$ are the interior zeros of ϕ_n , then ϕ_{n+1} has exactly one zero in each of*

$$(a, \xi_1), (\xi_1, \xi_2), \dots, (\xi_{n-2}, \xi_{n-1}), (\xi_{n-1}, b).$$

Proof. The “at least one” part is shown in the proof of Theorem 4.6. For uniqueness on each subinterval, note that the total number of interior zeros of ϕ_{n+1} is exactly n while there are exactly n subintervals listed; hence there can be at most one zero per subinterval. $\square$

Remark 4.8 (Wronskian sign correction). For $K(x) := p(\phi_n \phi'_{n+1} - \phi'_n \phi_{n+1})$,

$$K' = (\lambda_n - \lambda_{n+1}) r \phi_n \phi_{n+1}.$$

The sign is negative since $\lambda_n < \lambda_{n+1}$. Arguments using K should be phrased with this sign (monotonicity reversed relative to the mistaken $(\lambda_{n+1} - \lambda_n)$).

5. Rayleigh quotient, the first eigenvalue, and inequalities

Define the Rayleigh quotient on the admissible class (nonzero C^1 functions satisfying (1.3)):

$$R[y] := \frac{\int_a^b (py'^2 + qy^2) dx}{\int_a^b wy^2 dx}. \quad (5.1)$$

Lemma 5.1 (Green–Rayleigh identity). *If y is an eigenfunction with eigenvalue λ , then $R[y] = \lambda$.*

Proof. Multiply $L[y] = \lambda wy$ by y and integrate. By Lemma 1.2 and self-adjoint boundary conditions, the boundary term vanishes and we obtain $\int (py'^2 + qy^2) = \lambda \int wy^2$. $\square$

Lemma 5.2 (Elementary positivity of the first eigenfunction). *Let $p, q, w \in C([a, b])$ with $p, w > 0$ and consider the regular Sturm–Liouville problem with separated self-adjoint boundary conditions (1.3). Let λ_1 be the first eigenvalue given by the Rayleigh principle*

$$\lambda_1 = \inf_{0 \neq y \in \mathcal{D}} \frac{\int_a^b (p y'^2 + q y^2) dx}{\int_a^b w y^2 dx},$$

where $\mathcal{D}$ is the admissible class (functions satisfying (1.3)). Then any eigenfunction y_1 for λ_1 can be chosen strictly positive on (a, b) ; in particular, it has no interior zeros.

Proof. Step 1: pick a nonnegative minimizer. By the direct method (Arzelà–Ascoli and weak lower semicontinuity), there exists a minimizer y with $\int w y^2 = 1$ ¹. Replace y by $|y|$. We check that $|y| \in \mathcal{D}$: the boundary conditions are homogeneous and linear in $(y, p y')$, hence multiplying y by -1 near an endpoint preserves (1.3); if $y(a) = 0$ (resp. $y(b) = 0$) and $\sin \alpha \neq 0$ (resp. $\sin \beta \neq 0$), then necessarily $y'(a) = 0$ (resp. $y'(b) = 0$), so $|y|'$ satisfies the same endpoint relation. Moreover $|y|' = \operatorname{sgn}(y) y'$ a.e., hence

$$\int_a^b (p |y|'^2 + q |y|^2) dx = \int_a^b (p y'^2 + q y^2) dx, \quad \int_a^b w |y|^2 = \int_a^b w y^2.$$

Thus $|y|$ has the same Rayleigh quotient and is also a minimizer. Rename it $y_1 \geq 0$.

Step 2: Euler–Lagrange and regularity. Standard differentiation under variations within $\mathcal{D}$ (the boundary terms vanish because variations preserve (1.3)) yields the Euler–Lagrange equation

$$-(p y_1')' + q y_1 = \lambda_1 w y_1 \quad \text{on } (a, b),$$

so $y_1 \in C^2([a, b])$.

Step 3: no interior zero via the simplest ODE argument. Assume by contradiction that $y_1(x_0) = 0$ for some $x_0 \in (a, b)$. Since $y_1 \geq 0$, x_0 is a local minimum and hence $y_1'(x_0) = 0$. By the uniqueness lemma for second-order linear ODEs (the only solution of the IVP $y(x_0) = 0$, $y'(x_0) = 0$ is the zero solution), we must have $y_1 \equiv 0$, contradicting $\int w y_1^2 = 1$. Therefore $y_1 > 0$ on (a, b) . $\square$

Lemma 5.3 (From weak minimizer to classical C^2 eigenfunction). *Assume $p, q, w \in C([a, b])$ with $p > 0$, $w > 0$, and $p \in C^1([a, b])$. Let $y_1 \in H^1([a, b])$ be a (normalized) minimizer of the Rayleigh quotient under the separated self-adjoint boundary conditions (1.3). Then $y_1 \in C^2([a, b])$ and*

$$-(p y_1')' + q y_1 = \lambda_1 w y_1 \quad \text{pointwise on } (a, b).$$

Proof. Step 1: Weak Euler–Lagrange equation. For any φ in the admissible variation space (e.g. $C_c^\infty(a, b)$, or H^1 functions obeying the homogeneous boundary class), the first variation at y_1 gives

$$\int_a^b (p y_1' \varphi' + q y_1 \varphi) dx = \lambda_1 \int_a^b w y_1 \varphi dx.$$

Equivalently, for all $\varphi \in C_c^\infty(a, b)$,

$$-\int_a^b (p y_1')' \varphi dx + \int_a^b (q - \lambda_1 w) y_1 \varphi dx = 0,$$

so in the distribution sense

$$(p y_1')' = (q - \lambda_1 w) y_1 =: F.$$

¹For this course, we shall just assume the existence of this minimizer!

Step 2: Continuity of F and absolute continuity of py'_1 . Since $y_1 \in H^1([a, b])$, it has a continuous (indeed absolutely continuous) representative on $[a, b]$. With $q, w \in C$ we get $F \in C([a, b])$. A standard one-dimensional fact: if $G' = F \in L^1$ in the distribution sense on $[a, b]$, then G is absolutely continuous and $G(x) = G(a) + \int_a^x F$. Applying this to $G := py'_1$ yields

$$py'_1 \in AC([a, b]) \quad \text{and} \quad (py'_1)'(x) = F(x) \text{ for all } x.$$

Because F is continuous, $py'_1 \in C^1([a, b])$.

Step 3: Continuity of y'_1 and y''_1 . Since $p > 0$ and $p \in C^1$, we first get

$$y'_1 = \frac{py'_1}{p} \in C([a, b]).$$

Now expand

$$(py'_1)' = py''_1 + p'y'_1 = F \in C([a, b]).$$

Hence

$$y''_1 = \frac{F - p'y'_1}{p} \in C([a, b]),$$

because F, p', y'_1, p are continuous and $p > 0$. Thus $y_1 \in C^2([a, b])$, and the weak equation holds pointwise. $\square$

Theorem 5.4 (Existence and characterization of λ_1). *Let p, q, w be as in Section 1 and assume q is bounded below.² Then*

$$\lambda_1 = \inf_{0 \neq y \in \mathcal{D}} R[y], \quad (5.2)$$

and the infimum is attained by an eigenfunction y_1 which can be chosen strictly positive on (a, b) (hence has no interior zeros).

Proof. Normalize to $\int_a^b wy^2 = 1$ and consider a minimizing sequence $\{y_n\} \subset \mathcal{D}$ with $R[y_n] \downarrow \lambda_1$. Then $\int (py_n'^2 + qy_n^2)$ is bounded, hence $\int py_n'^2$ is bounded since q is bounded below. Because $[a, b]$ is compact and p is continuous and positive, there exists $c > 0$ with $p \geq c$, so $\int y_n'^2 \leq c^{-1} \int py_n'^2$ is bounded. By Cauchy–Schwarz, $|y_n(x) - y_n(s)| \leq \int_s^x |y'_n| \leq |x - s|^{1/2} \|y'_n\|_{L^2}$, so $\{y_n\}$ is equicontinuous. The normalization $\int wy_n^2 = 1$ and $w > 0$ imply a uniform L^2 bound, and hence (by equicontinuity) a uniform L^∞ bound. By Arzelà–Ascoli, there is a subsequence (not relabeled) with $y_n \rightarrow y$ uniformly and $y \in C^0[a, b]$. Since $\{y'_n\}$ is bounded in L^2 , after passing to a further subsequence we may assume $y'_n \rightharpoonup z$ weakly in L^2 . A standard result (e.g. [2, Ch. VIII]) shows $z = y'$ in the distribution sense and $y \in H^1$ with y' the weak limit.

Lower semicontinuity gives $\int_a^b py'^2 \leq \liminf_{n \rightarrow \infty} \int_a^b py_n'^2$. Continuity of q and uniform convergence yield $\int qy^2 = \lim_{n \rightarrow \infty} \int qy_n^2$. Likewise, $\int wy^2 = \lim_{n \rightarrow \infty} \int wy_n^2 = 1$. Therefore $R[y] \leq \liminf R[y_n] = \lambda_1$. Hence $R[y] = \lambda_1$ and $y \neq 0$ is a minimizer. A Lagrange multiplier variation of R under the normalization $\int wy^2 = 1$ yields the Euler–Lagrange equation $L[y] = \lambda_1 wy$ together with the boundary conditions (1.3) (the boundary terms vanish because variations preserve the boundary class). By Corollary ?? (or a maximum principle via Green’s function in Section ??), y has no interior zeros and can be chosen strictly positive. $\square$

Corollary 5.5 (Poincaré–Wirtinger type inequality). *For Dirichlet data $y(a) = y(b) = 0$ with $p = w \equiv 1$ and $q \geq 0$,*

$$\int_a^b y^2 dx \leq \frac{1}{\lambda_1} \int_a^b (y'^2 + qy^2) dx. \quad (5.3)$$

²If q is not bounded below, replace q by $q + Mw$ for a constant M large enough; this shifts all eigenvalues by M without changing eigenfunctions or the argument.

In particular, if $q \equiv 0$ then $\lambda_1 = \left(\frac{\pi}{b-a}\right)^2$ and

$$\int_a^b y^2 dx \leq \frac{(b-a)^2}{\pi^2} \int_a^b y'^2 dx. \quad (5.4)$$

Proof. Since $R[y] \geq \lambda_1$ for all admissible y , the first inequality is immediate. The Dirichlet Laplacian on (a, b) has $\lambda_1 = (\pi/(b-a))^2$ with eigenfunction $\sin(\pi(x-a)/(b-a))$. $\square$

5.1. Domain monotonicity (Dirichlet)

Proposition 5.6. *For Dirichlet boundary conditions, if $[c, d] \subset [a, b]$ then the first Dirichlet eigenvalue on $[c, d]$ satisfies $\lambda_1^{[c,d]} \geq \lambda_1^{[a,b]}$. More generally, $\lambda_n^{[c,d]} \geq \lambda_n^{[a,b]}$ for all n (min-max principle; see Remark 5.7).*

Proof. The admissible class for $[c, d]$ is contained in that for $[a, b]$ (extend functions by zero outside $[c, d]$), and Rayleigh's quotient cannot decrease upon restriction. The higher eigenvalue statement uses Courant–Fischer; see Remark 5.7. $\square$

Remark 5.7 (Advanced: higher eigenvalues). The Courant–Fischer min-max principle states $\lambda_n = \min_{\dim S=n} \max_{0 \neq y \in S} R[y]$, which implies all eigenvalue monotonicity/bracketing statements. A proof uses compact self-adjoint operator theory; see [4, Ch. VI] or [3, Ch. 3].

6. Liouville normal form and asymptotics

6.1. Liouville transform to Schrödinger form

Assume $p, q, w \in C^2([a, b])$ with $p > 0$ and $w > 0$. Define the monotone change of variable and scaling

$$t = t(x) := \int_{x_0}^x \sqrt{\frac{w(s)}{p(s)}} ds, \quad \kappa(x) := (p(x)w(x))^{1/4}, \quad u(t) := \kappa(x) y(x). \quad (6.1)$$

Then t maps $[a, b]$ bijectively onto $[\tau_a, \tau_b]$ with *Liouville length*

$$T := \tau_b - \tau_a = \int_a^b \sqrt{\frac{w}{p}} dx. \quad (6.2)$$

Write $\sigma := dt/dx = \sqrt{w/p}$ and $\rho := \sqrt{p/w} = 1/\sigma$, so that

$$\frac{d}{dt} = \rho \frac{d}{dx}, \quad \kappa_t = \rho \kappa_x.$$

Two identities (key). With $u = \kappa y$ one has

$$p y' = \kappa u_t - \kappa_t u, \quad (6.3)$$

and hence, differentiating w.r.t. x (use $d/dx = \sigma d/dt$),

$$(p y')' = \sigma (\kappa u_{tt} - \kappa_{tt} u). \quad (6.4)$$

Proof of (6.3): Since $y = u/\kappa$ and $y' = \sigma y_t$,

$$p y' = p \sigma y_t = \sqrt{pw} \frac{d}{dt} \left(\frac{u}{\kappa} \right) = \kappa^2 \left(\frac{u_t \kappa - u \kappa_t}{\kappa^2} \right) = \kappa u_t - \kappa_t u.$$

Then (6.4) follows by applying $d/dx = \sigma d/dt$ and canceling the middle terms.

Equation transform. Starting from the Sturm–Liouville equation

$$-(py')' + qy = \lambda w y,$$

use (6.4) to obtain

$$-\sigma(\kappa u_{tt} - \kappa_{tt}u) + q \frac{u}{\kappa} = \lambda w \frac{u}{\kappa}.$$

Multiply by κ/σ and note $w/\sigma = \sqrt{pw} = \kappa^2$ and $q/\sigma = (q/w)\kappa^2$ to get

$$-\kappa^2 u_{tt} + \kappa \kappa_{tt} u + \frac{q}{w} \kappa^2 u = \lambda \kappa^2 u.$$

Divide by κ^2 and set

$$-\frac{d^2 u}{dt^2} + V(t) u(t) = \lambda u(t), \quad t \in (\tau_a, \tau_b), \quad (6.5)$$

with

$$V(t) = \frac{q(x)}{w(x)} + \frac{\kappa_{tt}(t)}{\kappa(t)} \quad (x = x(t)). \quad (6.6)$$

This matches the standard Schrödinger form. A quick sanity check: if $p \equiv w \equiv 1$ then $\kappa \equiv 1$, $t = x$, and $V \equiv q$.

Boundary conditions (explicit map). A separated self-adjoint condition at $x = a$,

$$y(a) \cos \alpha + p(a) y'(a) \sin \alpha = 0,$$

becomes a Robin condition for u at $t = \tau_a$. Using (6.3) and multiplying by $\kappa(a)$,

$$\left(\cos \alpha - \kappa(a) \kappa_t(a) \sin \alpha \right) u(\tau_a) + \kappa(a)^2 \sin \alpha u_t(\tau_a) = 0.$$

Since $\kappa_t = \rho \kappa_x$ and $\kappa_x/\kappa = \frac{1}{4} \left(\frac{p'}{p} + \frac{w'}{w} \right)$, we have

$$\kappa(a) \kappa_t(a) = \frac{1}{4} \left(p'(a) + p(a) \frac{w'(a)}{w(a)} \right), \quad \kappa(a)^2 = \sqrt{p(a)w(a)}.$$

Thus the mapped coefficients can also be written as

$$\left(\cos \alpha - \frac{1}{4} (p' + p w'/w) \sin \alpha \right) u(\tau_a) + \sqrt{pw} \sin \alpha u_t(\tau_a) = 0 \quad (\text{evaluated at } x = a).$$

An analogous formula holds at $x = b$ with β in place of α . These are separated, real Robin data for $-d^2/dt^2 + V$, hence self-adjoint. Zeros are preserved because $\kappa > 0$ and $x \mapsto t$ is strictly increasing.

Unitarity of the transform (useful check). The map $U : y \mapsto u = \kappa y$ is an isometry

$$\int_a^b |y|^2 w dx = \int_{\tau_a}^{\tau_b} |u|^2 dt \quad (\text{since } w/\kappa^2 = \sqrt{w/p} = \sigma, \quad dt = \sigma dx).$$

So U identifies $L_w^2([a, b]; dx)$ with $L^2([\tau_a, \tau_b]; dt)$ and conjugates the formal Sturm–Liouville expression to the Schrödinger operator in (6.5).

6.2. Asymptotic eigenvalue spacing

Theorem 6.1 (Weyl-type asymptotics in 1D). *For a regular Sturm–Liouville problem with continuous p, q, w , the eigenvalues satisfy*

$$\lambda_n \sim \left(\frac{n\pi}{T}\right)^2 \quad (n \rightarrow \infty), \quad (6.7)$$

where T is the Liouville length (6.2).

Proof (elementary sandwich). After the Liouville transform, we obtain the Schrödinger operator $-u'' + Vu$ on (τ_a, τ_b) with $V \in C([\tau_a, \tau_b])$. Let $m := \min V$ and $M := \max V$. For Dirichlet boundary conditions (the other separated cases are analogous), comparison of Rayleigh quotients gives

$$m + \left(\frac{n\pi}{T}\right)^2 \leq \lambda_n \leq M + \left(\frac{n\pi}{T}\right)^2. \quad (6.8)$$

(Upper bound: test against $\sin\left(\frac{n\pi(t-\tau_a)}{T}\right)$ in the Rayleigh quotient for $-u'' + Vu$; lower bound: replace V by m in the quadratic form.) Subtract $(n\pi/T)^2$ and let $n \rightarrow \infty$ to obtain (6.7). $\square$

Remark 6.2. Sharper asymptotics (e.g. next-order terms via WKB/Prüfer) are available under higher regularity; see [3, Ch. 2] or [6, §9.3]. We will not need them here.

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